UNIVERSITY OF OULU

Project Report | Affective Computing

When Voice and Face Disagree: Analysing Modality Conflict in Audio-Visual Emotion Recognition

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Abstract

People express emotions through both the face and the voice, but these two channels do not always show the same emotion. This study examines when and why an audio model and a visual model disagree in emotion recognition. Using the CREMA-D dataset, we trained two ResNet18 models with the same design, one on speech Mel-spectrograms and one on face images, and evaluated them on 19 speakers not seen during training. The two models were almost equally accurate (67.8% for audio and 68.0% for video), but they disagreed on 51% of the test clips. Comparison with human ratings showed that 47% of these disagreements matched a real voice–face conflict perceived by humans; the rest were model errors. The face was more useful for happiness and disgust, and the voice for sadness, neutral and anger. Recognition patterns differed strongly between individual speakers and were not clearly explained by gender or age. A fusion method that uses the models' confidence and their disagreement improved accuracy to 81.8%, higher than human raters who both saw and heard the clips (76.5%). These results show that disagreement between voice and face is a useful signal for multimodal emotion recognition.

Keywords: emotion recognition; multimodal fusion; modality conflict; CREMA-D; human perception

Abbreviations: CNN, convolutional neural network; CREMA-D, Crowd-sourced Emotional Multimodal Actors Dataset; ECE, expected calibration error; RQ, research question

1. Introduction

People show their emotions in more than one way. Two of the most important ways are the face and the voice. Computers can learn to recognise emotions from speech (1) and from facial expressions (2). Many studies combine the two, because using both often gives better results than using only one (3, 4).

Most of these studies implicitly assume that the face and the voice show the same emotion. This is not always true. Research on human perception shows that people use both the face and the voice when they judge emotions. When the two do not match, people's judgements change (5). For example, a person may control their face but still sound upset (6). Multimodal computer models face a related problem: when two modalities are trained together, one of them can dominate and the other is used less (7, 8). Some recent studies have started to look at these mismatches (6, 9).

These studies mainly propose new models that handle mismatches better. They do not analyse in detail **when and why separately trained audio and visual models disagree**: which emotions and which speakers cause disagreement, and whether a disagreement means that the face and the voice really show different emotions, or just that one model made a mistake. It is also not clear whether information about disagreement can help a system make better decisions.

To study this, we used the CREMA-D dataset (10). This dataset is useful because people rated each clip three times: by listening only, by watching only, and by both watching and listening. We trained two models with the same design, ResNet18 (11): one for audio and one for video. Using the same design for both makes the comparison fair. We then answered five research questions:

- **RQ1:** How often do the audio and visual models disagree?
- **RQ2:** Which emotions are harder to recognise from audio or from video?
- **RQ3:** Which modality is more useful for each emotion?
- **RQ4:** Do different speakers show emotions in different ways?
- **RQ5:** Can information about disagreement improve emotion recognition?

Our main findings are as follows. The two models disagreed on about half of the test clips. The face worked better for happiness and disgust, while the voice worked

Table 1: Speaker-independent split of CREMA-D.

Set	Clips	Speakers	Men / Women
Training	4,667	57	30 / 27
Validation	1,223	15	8 / 7
Test	1,552	19	10 / 9
Total	7,442	91	48 / 43

better for sadness, neutral and anger. The better modality also changed from speaker to speaker. In about 47% of model disagreements, human raters also saw a real difference between the face and the voice. Finally, combining the two models in a way that uses their confidence and their disagreement increased accuracy from 68.0% to 81.8%.

2. Materials and Methods

2.1 Dataset

We used CREMA-D (10), which contains 7,442 video clips of 91 actors (48 men and 43 women). Each actor said 12 short sentences while showing one of six emotions: anger, disgust, fear, happiness, neutral and sadness. One sentence ("It's eleven o'clock") was recorded at three intensity levels (low, medium and high); all other sentences have no intensity label. The emotions are well balanced: each emotion makes up about 17% of the clips, except neutral with about 15%. For each clip, the dataset gives the emotion the actor was asked to show. It also gives the emotions that human raters saw when they only listened, only watched, or both watched and listened (10).

We split the data by speaker, so that no actor appeared in more than one part. This is important because a model that has seen a speaker during training could recognise that person's voice or face instead of their emotion. First, 20% of the speakers were randomly chosen for the test set (random seed 42), keeping the same share of men and women. From the remaining speakers, 20% were then kept for validation in the same way. Table 1 shows the resulting split. Both models used exactly the same split. We removed one validation clip (`1076_MTI_SAD_XX`) because its audio was silent and its video showed no face, leaving 1,222 validation clips.

2.2 Audio model

We turned each audio recording into a Mel-spectrogram, an image that shows how the energy of the sound changes over time and frequency. Each spectrogram was resized to 224×224 pixels so that it could be used as input to ResNet18 (11), a network pretrained on the ImageNet image dataset. We replaced the last layer of the network with a new layer for our six emotions and trained the whole network on our data.

A first version of the audio model, trained without data augmentation, overfitted strongly (Section 3.1). We therefore trained the final audio model with several common techniques that make it harder for a network to memorise the training examples: *SpecAugment*, which hides random frequency bands and time segments of the spectrogram; random shifts in time; *mixup*, which trains on mixtures of two examples; *label smoothing*; and dropout before the last layer. The choice between the two versions was made on the validation set only.

2.3 Visual model

From each video, we took eight frames spread evenly over the clip. In each frame, we found the face with the OpenCV face detector, cut it out and resized it to 224×224 pixels. If no face was found in a frame, we used the face position from the nearest frame. Each face image was passed through ResNet18, which gave a probability for each emotion. We then averaged the probabilities of the eight frames to get one prediction for the whole video.

For both models, we chose the final version using only the validation set. Table 2 gives the training details, and Figure 1 shows an overview of the study.

2.4 Combining the two models

We combined the two models after prediction, using only the emotion probabilities from each model. Neural networks are often too confident in their predictions (12). We therefore also tested *temperature scaling* (12), a simple method that makes the probabilities match how often the model is actually right.

Each model has only one output: a probability for each of the six emotions (the softmax output of the network). There is no separate confidence output. We define the *confidence* of a model as its highest probability, that is, the probability of the emotion it predicts. We tested several ways of combining the two models:

- **Average or product:** combine the probabilities of the two models directly.
- **Most confident model:** use the prediction of the model that is more confident.
- **Weighted average:** give each model a weight:

$$p_{\text{combined}} = w \cdot p_{\text{visual}} + (1 - w) \cdot p_{\text{audio}} \tag{1}$$

- **Stacking:** train a small multinomial logistic regression model that learns which model to trust in each situation. For each clip, its input is a vector of 16 numbers, all computed from the two (temperature-scaled) probability vectors p_{visual} and p_{audio}:

$$x = \big[\, \log p_{\text{visual}},\ \log p_{\text{audio}},\ c_{\text{visual}},\ c_{\text{audio}},\ c_{\text{visual}} - c_{\text{audio}},\ d \,\big] \tag{2}$$

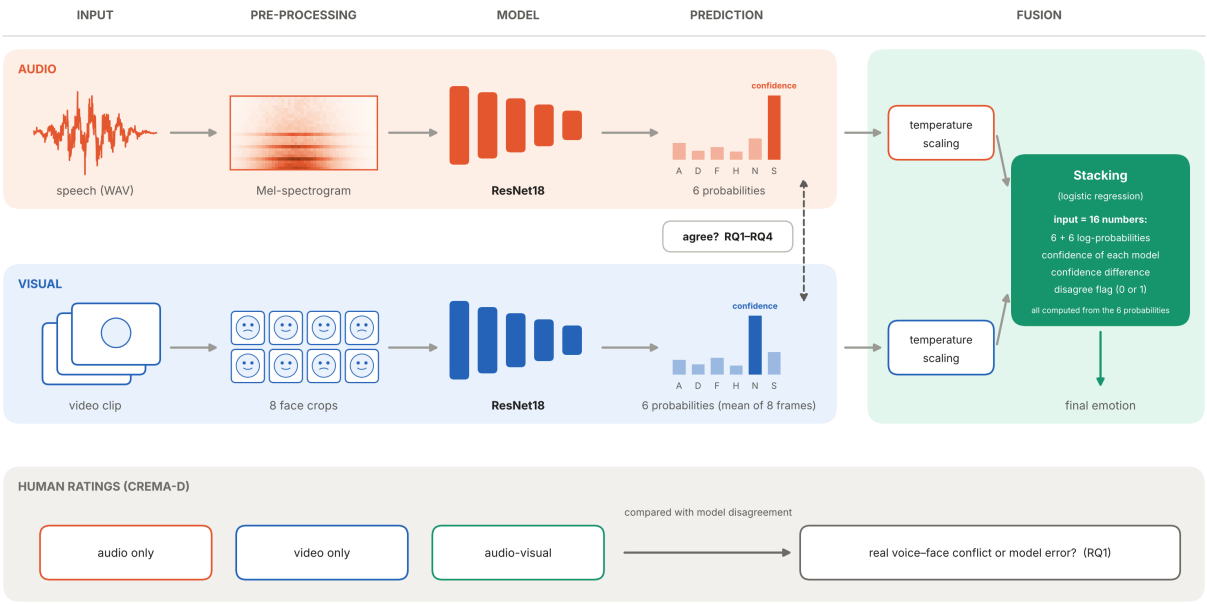


Figure 1: Overview of the study. Two ResNet18 models predict emotions from audio and from face images. Each model has only one output: six emotion probabilities. The *confidence* of a model is its highest probability; it is not a separate output. The agreement of the two models is analysed (RQ1–RQ4) and compared with human ratings. For fusion (RQ5), a stacking model receives 16 numbers, all computed from the two sets of probabilities (Equation 2).

Table 2: Training settings of the two models.

	Audio model	Visual model
Input	Mel-spectrogram (16 kHz, 128 Mel bands, 25 ms window, 10 ms hop)	8 face images per video
Image size	224×224	224×224
Network	ResNet18, pretrained on ImageNet	ResNet18, pretrained on ImageNet
Optimiser	AdamW, weight decay 10^{-3}	AdamW, weight decay 10^{-4}
Learning rate (backbone / last layer)	10^{-4} / 10^{-3}, cosine decay	10^{-4} / 10^{-3}, then 10^{-5} / 10^{-4}
Batch size	32 spectrograms	8 videos (64 images)
Data augmentation	SpecAugment, random time shift, mixup	Horizontal flip
Other regularisation	Label smoothing (0.1), dropout (0.3)	–
Early stopping	After 8 epochs without improvement (max. 40)	After 3 epochs without improvement
Selected model	Epoch 33	Second training stage, epoch 4
Validation macro-F1	0.666	0.647

Here, the first two terms are the six log-probabilities of each model (12 numbers), c_{visual} and c_{audio} are the confidences of the two models, and d is 1 if the two models predict different emotions and 0 otherwise. The output is a probability for each of the six emotions.

Everything that had to be learned (the temperatures, the weight w and the stacking model) was learned on the validation set only. The final results are reported once on the test set.

2.5 Human ratings

For each clip, we used the emotion chosen most often by human raters in each of the three conditions: audio only, video only, and audio-visual. When two emotions were chosen equally often, both were kept. We said that humans saw a *real conflict* between the face and the voice when the audio-only and video-only ratings were different. This allowed us to check whether model disagreements are real conflicts or model mistakes.

2.6 Evaluation

Performance measures. We used the following measures to describe how well the models worked:

- **Accuracy:** the share of clips for which the predicted emotion was correct.
- **Precision and recall:** for one emotion, precision is the share of clips predicted as that emotion that were really that emotion. Recall is the share of clips of that emotion that the model found.
- **F1-score:** a single value that combines precision and recall (their harmonic mean). It is high only when both are high.
- **Macro-F1:** the average F1-score of the six emotions. Unlike accuracy, it gives every emotion the same weight.
- **Confusion matrix:** a table that shows, for each true emotion, how often it was predicted as each of the six emotions. It shows which emotions are mixed up.
- **Expected calibration error (ECE):** the average difference between how confident a model is and how often it is actually right. A value of 0 means the confidence can be fully trusted.

Statistical tests. To check whether a difference is real and not due to chance, we used the following tests. A difference was considered significant when $p < 0.05$.

- **McNemar's test** compares two models on the same clips. It looks at the clips where only one of the two models was right and checks whether one model wins clearly more often.
- **Bootstrap confidence intervals** show how certain a result is. We repeatedly drew a random sample of speakers (2,000 times), computed the result for each sample, and took the range that contains 95% of the results. We sampled speakers rather than clips because clips from the same speaker are not independent.
- **Chi-square (χ^2) test** checks whether two things are related, for example whether accuracy differs between speakers more than expected by chance.
- **Mann–Whitney U test** compares two groups of speakers, such as women and men.
- **Wilcoxon signed-rank test** compares the two models over the same speakers.
- **Spearman correlation (ρ)** measures whether two values tend to rise together, for example a speaker's visual and audio accuracy. It ranges from -1 to 1, where 0 means no relation.
- **Holm correction:** when many tests are done at once, some can look significant by chance. Holm's method adjusts the p-values to prevent this. We used it when comparing the two models for the six emotions.

3. Results

3.1 Performance of the two models

3.1.1 Training behaviour

Figure 2 shows how accuracy and loss changed during training for both models.

Audio model. The first version of the audio model, trained without data augmentation, reached almost 100% training accuracy after only ten epochs, while its validation accuracy stayed below 59% and its validation loss rose from 1.24 to about 2.0. In other words, it *overfitted*: it partly memorised the training examples instead of learning general patterns. It was also strongly *overconfident*: on the validation set, its average confidence was 86% while it was right for only 59% of the clips (ECE 0.27). The final audio model, trained with data augmentation, learned more slowly and more steadily (Figure 2a, c). Its validation loss fell from 1.48 to about 1.0 and stayed there, and its validation accuracy kept improving until epoch 33 (66.8%, macro-F1 0.666). Its training accuracy still approached 100%, so some overfitting remains, which is common when training data are limited, as in CREMA-D with 57 training speakers. However, unlike in the first version, this no longer harmed the validation results. All results below use this final audio model.

Visual model. The visual model learned more slowly and overfitted less. In the first stage, the best validation result was reached at epoch 3, after which the validation loss started to increase. We therefore started a second training stage from epoch 3 with a ten times smaller learning rate (grey area in Figure 2b, d). This lowered the validation loss to about 1.3 and raised validation accuracy to 64.6%. Training accuracy at the selected epoch was 89%, which still shows some overfitting.

3.1.2 Results on the test set

Table 3 compares validation and test results. Both models reached similar or slightly better results on the test set than on the validation set. This shows that our choice of model on the validation set did not lead to overly optimistic results, and that the models work on speakers they have never seen.

On the 1,552 test clips, the visual model was correct for 1,056 clips (68.0%) and the audio model for 1,052 clips (67.8%). Both models are far better than random guessing, which would give about 17% for six emotions. Overall, the two models were almost equally accurate.

Table 4 gives the results for each emotion. The similar overall accuracy hides large differences between emotions. The visual model recognised almost all happiness

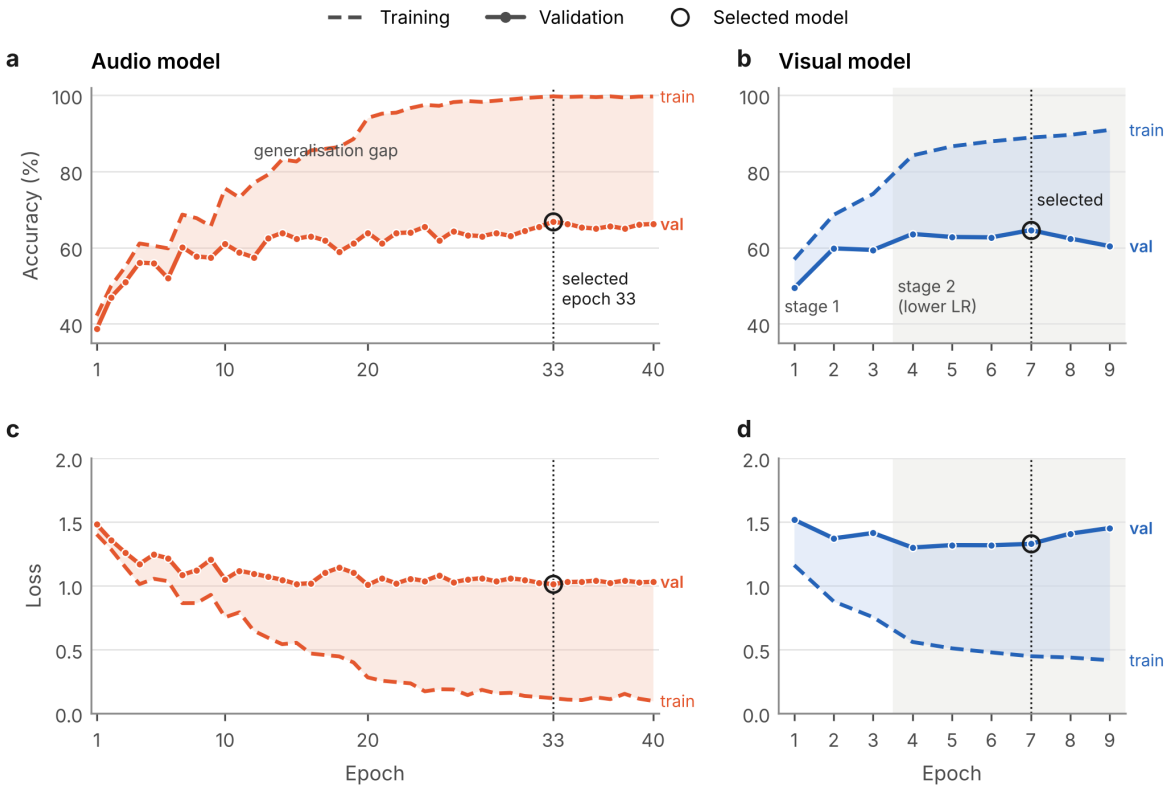


Figure 2: Accuracy (a, b) and loss (c, d) on the training and validation sets during training of the final models. The shaded area between the two curves shows the gap between training and validation performance. Circles and dotted lines mark the selected models. For the visual model, the grey background marks the second training stage, which started from the best model of the first stage (epoch 3).

Table 3: Validation and test results of the two models.

	Validation		Test	
	Acc. (%)	Macro-F1	Acc. (%)	Macro-F1
Audio	66.8	0.666	67.8	0.677
Visual	64.6	0.647	68.0	0.673

clips (95.8%) but only about a third of the sadness clips (36.2%). The audio model was more balanced: its recall ranged from 55.1% (disgust) to 87.2% (neutral). These differences are studied in detail in Sections 3.3 and 3.4.

3.1.3 Which emotions are confused?

The confusion matrices in Figure 3 show which emotions each model mixed up. Each row shows how the clips of one true emotion were classified.

- **Audio model:** the most common errors were disgust and fear clips classified as sadness (22% and 17%), sadness clips classified as neutral (15%), and happiness clips classified as anger (12%). Sadness, disgust and fear are all low-energy emotions in the voice, while happiness and anger are both loud and high-energy. This may explain why the audio model mixed them up.
- **Visual model:** most errors concerned sadness. Only 36% of sadness clips were recognised. The rest were mostly classified as neutral (23%), fear (17%) or disgust (13%). A sad face is often subtle and can look similar to a neutral face. Neutral clips were also confused with anger, fear and sadness (10–12% each). In contrast, happiness was almost never confused with another emotion (96% correct), probably because a

Table 4: Test results of the two models for each emotion. P = precision, R = recall. All values are in %.

		Audio			Visual		
Emotion	n	P	R	F1	P	R	F1
Anger	265	74.0	76.2	75.1	70.6	63.4	66.8
Disgust	265	68.9	55.1	61.2	76.8	82.3	79.4
Fear	265	71.4	59.2	64.7	57.8	66.0	61.6
Happiness	265	71.6	65.7	68.5	95.5	95.8	95.7
Neutral	227	65.1	87.2	74.6	53.9	63.9	58.5
Sadness	265	58.3	66.0	61.9	50.0	36.2	42.0
Accuracy	1,552		67.8			68.0	
Macro-F1			67.7			67.3	

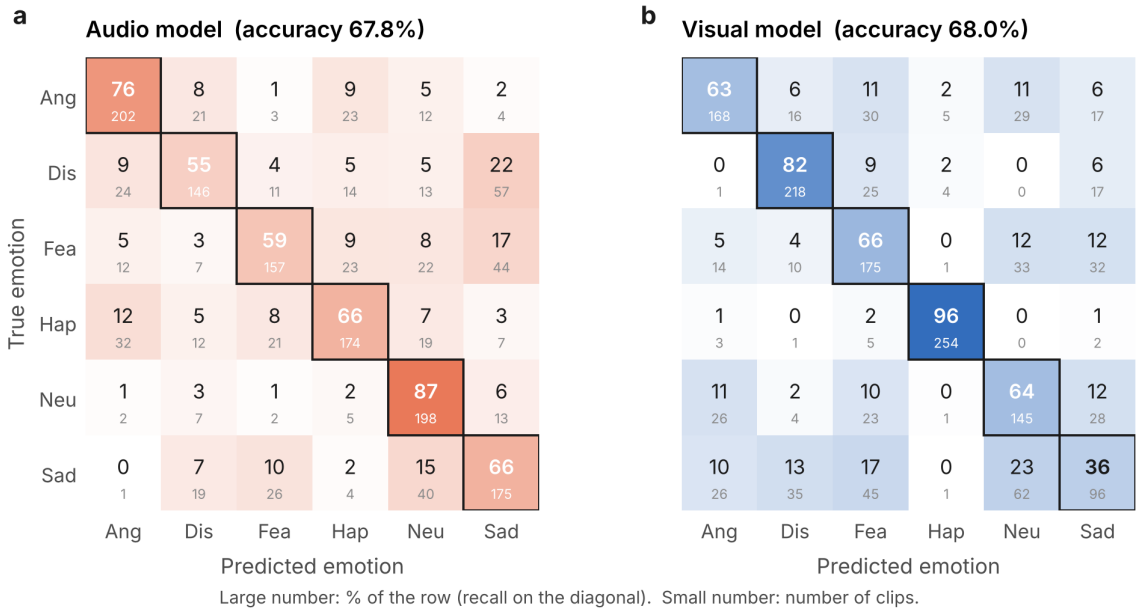


Figure 3: Confusion matrices of (a) the audio model and (b) the visual model on the test set. Each row is a true emotion and each column a predicted emotion. The large number in each cell is the percentage of the row; the small number is the number of clips. The diagonal (framed cells) shows the correctly recognised clips.

smile is a very clear facial signal.

3.2 How often do the two models disagree? (RQ1)

3.2.1 Overall disagreement

The two models predicted different emotions for 795 of the 1,552 test clips (**51.2%**). They agreed on the remaining 757 clips (48.8%). In other words, the audio and visual models disagreed on about half of the test set, even though they were almost equally accurate.

Table 5 shows how agreement and correctness are related. When the two models agreed, they were almost always right: 717 of the 757 agreements (94.7%) were correct. Only 40 clips received the same wrong label from both models. When the models disagreed, the visual model was right in 339 cases (42.6% of the disagreements), the audio model in 335 cases (42.1%), and neither model in 121 cases (15.2%). So agreement is a strong sign of a correct prediction, and disagreement is a strong sign that at least one model is wrong.

3.2.2 Disagreement by emotion and intensity

Disagreement was not the same for all emotions (Figure 4a). It was highest for sadness (74.7%) and lowest for happiness (36.2%). The models also disagreed more when the emotion was acted weakly. For the one sentence recorded at three intensity levels, disagreement fell from 72.6% at low intensity to 51.6% at medium and 44.2% at high intensity (95 clips each).

Table 5: Agreement and correctness of the two models on the test set.

Outcome	Clips	Share (%)
Models agree	**757**	**48.8**
Both correct	717	46.2
Both wrong (same label)	40	2.6
Models disagree	**795**	**51.2**
Only visual correct	339	21.8
Only audio correct	335	21.6
Both wrong (different labels)	121	7.8
Total	1,552	100.0

3.2.3 Is model disagreement a real conflict between voice and face?

A disagreement between two models can have two causes. Either the voice and the face really show different emotions, or one of the models simply made a mistake. For example, if an actor sounds sad but looks neutral, a model disagreement reflects a real conflict; if the actor both sounds and looks sad, a disagreement can only be a model error. To separate these two cases, we used the human ratings in CREMA-D.

Human raters also often disagreed between the two channels. On 658 test clips (**42.4%**), the emotion chosen by people who only listened was different from the one chosen by people who only watched. So voice–face conflict is common, even for human perception.

Model disagreement was related to human disagree-

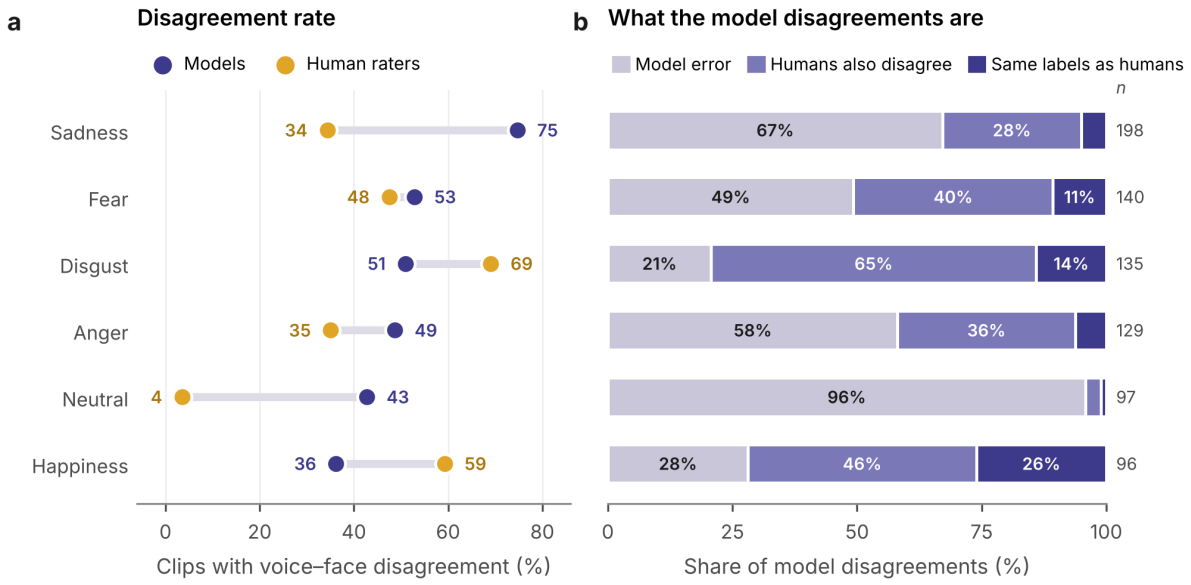


Figure 4: (a) Share of test clips on which the two models (purple) and the two groups of human raters (yellow) chose different emotions from the voice and from the face, for each emotion. (b) What the model disagreements are. *Model error*: humans saw the same emotion in voice and face, so one model was wrong. *Humans also disagree*: humans also saw different emotions in voice and face. *Same labels as humans*: each model gave the same label as the human raters of its own channel. n = number of model disagreements.

ment, but far from identical. When humans saw a conflict, the models disagreed in 56.2% of the clips. When humans saw no conflict, the models still disagreed in 47.5% of the clips (χ^2 test, $p < 0.001$). Of the 795 model disagreements, 370 (46.5%) happened on clips where humans also disagreed between voice and face. The other 425 (53.5%) happened on clips where humans saw the same emotion in both channels, so one of the models must have made a mistake. In only 78 clips (9.8%) did both models give exactly the same labels as the human raters of their own channel.

Figure 4b shows that this depends strongly on the emotion. For neutral clips, almost all model disagreements (96%) were model errors, because humans almost never disagreed on neutral clips (3.5%). For sadness, two thirds (67%) were model errors. In contrast, for disgust and happiness, most model disagreements (79% and 72%) happened on clips where humans also disagreed. Here, people who only listened often did not hear the emotion that people who only watched could see. Figure 4a also shows that the emotions that caused the most disagreement were different for models and humans. Humans disagreed most on disgust (69.1%) and happiness (59.2%), while the models disagreed most on sadness (74.7%).

3.2.4 Comparison with human accuracy

Table 6 compares the models with human raters who received the same information. The audio model was clearly better than humans who only listened (67.8% vs 47.9%). The visual model was about as good as humans who only watched (68.0% vs 69.5%). When both channels were combined, our best fusion model (Section 3.6) was better than humans who both watched and listened (81.8% vs 76.5%; McNemar test, $p < 0.001$). This shows that recognising emotions from these clips is difficult for people too, and that the models reach a human level of performance.

Table 6: Accuracy (%) of human raters and of our models when given the same information. A human rating was counted as correct when the intended emotion was the most frequently chosen label.

Information	Humans	Our models
Audio only	47.9	67.8 (audio model)
Video only	69.5	68.0 (visual model)
Audio and video	76.5	81.8 (fusion)

3.3 Which emotions are harder to recognise? (RQ2)

The difficulty of an emotion depended strongly on the modality (Table 4).

Audio model. The hardest emotions for the audio model were disgust (F1 = 61.2%) and sadness (61.9%). Only 55.1% of the disgust clips were recognised. The easiest emotions were anger (F1 = 75.1%) and neutral (74.6%). In general, the audio model performed similarly across emotions: the difference between its best and worst emotion was 14 percentage points.

Visual model. The visual model depended more on the emotion, with a difference of 54 percentage points between its best and worst emotion. The hardest emotion

by far was sadness (F1 = 42.0%). Only 96 of the 265 sadness clips were recognised from the face (recall 36.2%). Neutral was the second hardest emotion (F1 = 58.5%). In contrast, happiness (95.7%) and disgust (79.4%) were recognised very well.

Hard for both models. Some clips were difficult for both models. The share of clips that neither model recognised was highest for sadness (20.0%) and fear (16.6%), and lowest for happiness (1.9%). Sadness and fear are therefore the most difficult emotions overall, while happiness is the easiest.

Comparison with human raters. Figure 5 compares the models with human raters who received the same information. For the face, humans showed a similar pattern to the visual model. Sadness was also the hardest emotion for people who only watched (30.2%), and happiness the easiest (97.0%). This suggests that the difficulty of sadness lies mainly in how it is expressed: sadness was often hard to see in the actors' faces. For the voice, humans found most emotions hard to hear. Their recall was below 40% for sadness, disgust, happiness and fear. The audio model was better than these human raters for every emotion except neutral. Humans recognised almost all neutral clips (98% from the voice and 99% from the face), probably because people tend to choose "neutral" when they are unsure.

3.4 Which modality is more useful for each emotion? (RQ3)

No single modality was the best for all emotions. Figure 6a shows the difference in F1-score between the visual and the audio model for each emotion, with 95% confidence intervals calculated over speakers.

- **The face was more useful for happiness and disgust.** The visual model was much better for happiness (+27.2 points) and disgust (+18.2 points).
- **The voice was more useful for sadness and neutral.** The audio model was 19.9 points better for sadness and 16.1 points better for neutral.
- **For anger, the voice was somewhat better.** The audio model recognised significantly more anger clips (recall 76.2% vs 63.4%), but the difference in F1 (8.3 points) was not significant over speakers.
- **For fear, neither modality was clearly better.** The visual model found slightly more fear clips, but the audio model was more precise, and the difference was not significant (Table 7).

Who is right when the models disagree? A direct way to compare the two modalities is to look only at the 795 clips on which the models disagreed (Figure 6b). For happiness, the face was right in 90% of these clips and the voice in only 6%. For disgust, the face was also usually right (71%). For neutral, the opposite was true: the voice was right in 73% of the disagreements and the face in only 19%. The voice was also usually right for sadness (59%) and anger (56%). For fear, the two modalities were right about equally often (46% vs 33%).

Table 7: Comparison of the two modalities for each emotion. p: McNemar test on the clips of that emotion, Holm-corrected for six tests. Best: F1 of the better single model. Fusion: weighted average (Section 3.6). F1 values in %.

Emotion	Better modality	p	Best	Fusion	Change
Happiness	Visual	< 0.001	95.7	94.7	-1.0
Disgust	Visual	< 0.001	79.4	81.6	$+2.2$
Fear	Neither	0.11	64.7	75.8	$+11.1$
Anger	Audio (recall)	0.003	75.1	85.8	$+10.7$
Neutral	Audio	< 0.001	74.6	78.2	$+3.6$
Sadness	Audio	< 0.001	61.9	62.9	$+1.0$

Table 8: Accuracy (%) by emotion intensity (sentence "It's eleven o'clock", 95 test clips per level).

Intensity	Audio model	Visual model
Low	50.5	55.8
Medium	65.3	64.2
High	68.4	78.9

The two modalities complement each other. Table 7 also shows what happens when the two models are combined (weighted fusion, Section 3.6). For emotions where one modality was clearly better (happiness, disgust, neutral and sadness), combining the models changed the F1-score by less than 4 points. Here, the second modality had little to add. For anger and fear, where the F1 difference between the modalities was not significant, combining the models improved the F1-score by about 11 points. This shows that the voice and the face carry different and useful information mainly for these two emotions.

The effect of emotion intensity. One sentence in CREMA-D was recorded at three intensity levels: low, medium and high (95 test clips each). As Table 8 shows, both models improved when emotions were expressed more strongly, but the face benefited more: the visual model improved from 55.8% at low intensity to 78.9% at high intensity, and the audio model from 50.5% to 68.4%. At low intensity, the two modalities were about equally good. Because each level contains only 95 clips, these numbers should be read as trends.

3.5 Do different speakers show emotions in different ways? (RQ4)

3.5.1 Speakers differ in how easy they are to recognise

Recognition accuracy varied between the 19 test speakers (Figure 7a). For the visual model, accuracy ranged from

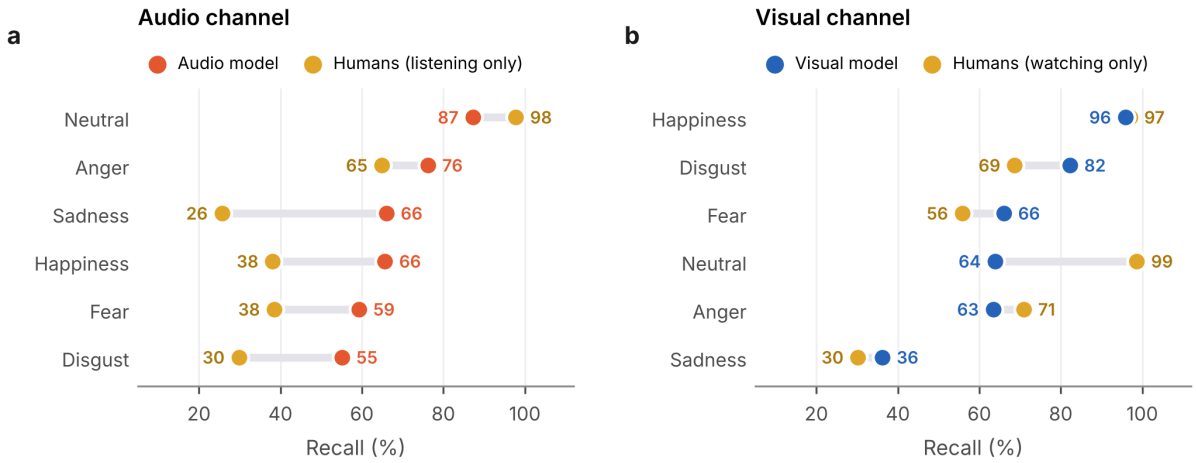


Figure 5: Recall for each emotion of (a) the audio model and humans who only listened, and (b) the visual model and humans who only watched. Emotions are sorted by the recall of the model.

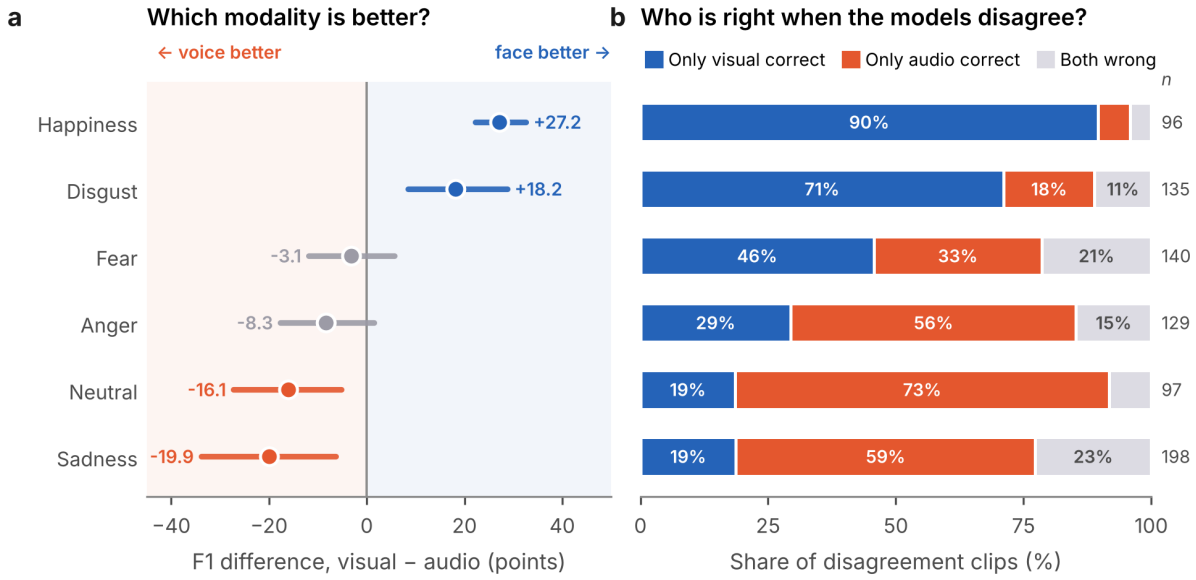


Figure 6: (a) Difference in F1-score between the visual and the audio model for each emotion. Lines show 95% confidence intervals over speakers. Coloured points show significant differences; grey points show differences that are not significant. (b) For the clips on which the two models disagreed: the share where only the visual model, only the audio model, or neither model was correct. n = number of disagreement clips.

46.3% to 87.8%. For the audio model, it ranged from 56.1% to 79.3%. These differences between speakers were statistically significant for both models (χ^2 test; visual $p < 0.001$, audio $p = 0.035$). The visual model varied much more between speakers (standard deviation 12.4 points) than the audio model (6.7 points). How often the two models disagreed also depended on the speaker, from 37.8% to 68.3% of a speaker's clips ($p = 0.005$).

3.5.2 Some speakers express emotions more clearly in the face, others in the voice

A speaker who was easy to recognise from the face was not necessarily easy to recognise from the voice. The visual and audio accuracies of the speakers were unrelated (Spearman $\rho = 0.06$, $p = 0.81$). For example, speaker 1025 was recognised very well from the face (86.6%) but much less well from the voice (63.4%). Speaker 1037 showed the opposite pattern: 75.6% from the voice but only 54.9% from the face.

The voice gave better results for 12 of the 19 speakers, the face for 6, and one speaker was recognised equally well by both. Over all speakers, neither modality was better (Wilcoxon test, $p = 0.97$). For 6 speakers, the difference between the two models was statistically significant on its own (McNemar test, $p < 0.05$, marked with * in Figure 7a): 4 speakers were recognised better from the face and 2

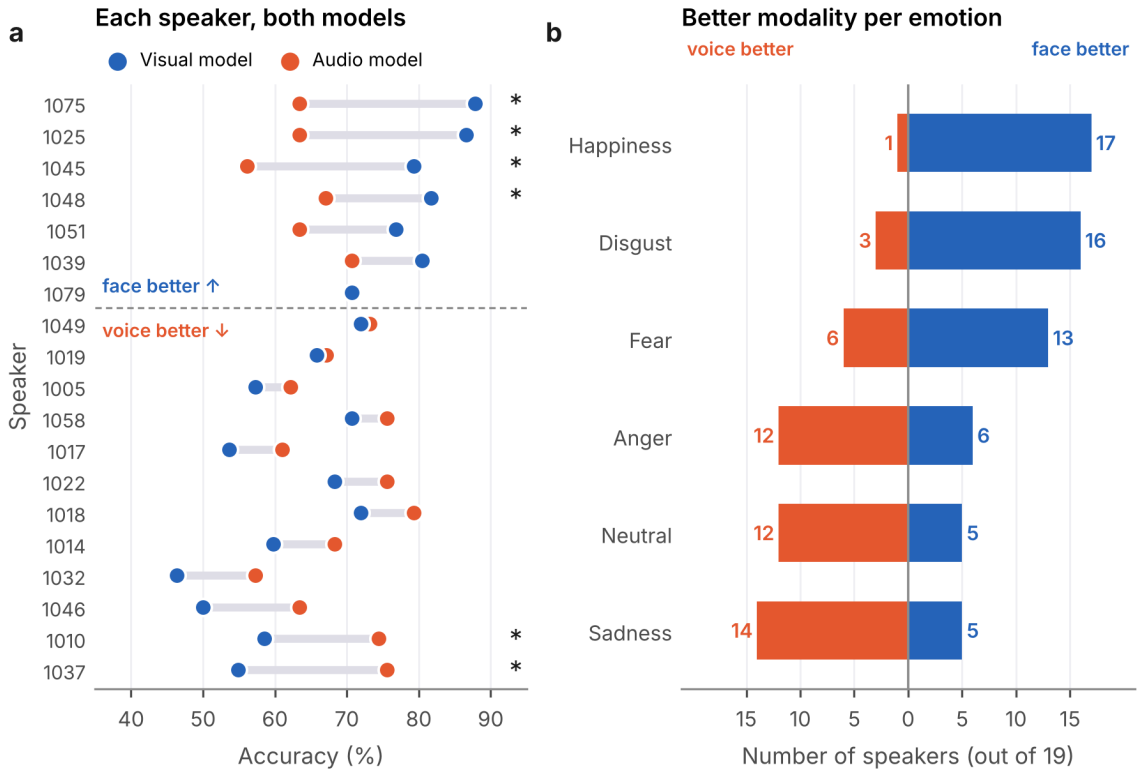


Figure 7: (a) Accuracy of the visual and audio models for each of the 19 test speakers, sorted by the difference between the two models. Speakers above the dashed line were recognised better from the face (speaker 1079 equally well), those below better from the voice. * = significant difference (McNemar test, $p < 0.05$). (b) For each emotion, the number of speakers for whom the face (blue) or the voice (orange) gave a higher recall.

better from the voice. For the other 13 speakers, the two modalities worked about equally well.

3.5.3 Which patterns are shared by all speakers?

Figure 7b shows, for each emotion, for how many speakers the face or the voice worked better. Some patterns were the same for almost everyone. Happiness was recognised better from the face for 17 of the 19 speakers, and disgust for 16. Sadness was recognised better from the voice for 14 speakers, and neutral for 12. Other emotions depended more on the speaker. For anger, the voice was better for 12 speakers but the face for 6; for example, anger recall was 85 points higher from the voice for speaker 1019, but 50 points higher from the face for speaker 1025. Fear was recognised better from the face for 13 speakers and from the voice for 6. This explains why anger and fear had no clearly better modality in F1 (Section 3.4): the answer partly depends on the person.

Because the better modality changes from speaker to speaker, combining the two models should help most speakers. Indeed, the weighted fusion (Section 3.6) gave a higher accuracy than the speaker's better single model for 16 of the 19 speakers. For the other three speakers, fusion was equal to or only slightly below (by 1.2 points) the better single model.

3.5.4 Gender and age

We also tested whether the differences between speakers could be explained by gender or age (Table 9). For the visual model and for the disagreement rate, women and men were very similar. For the audio model and for the fused model, women tended to be recognised better (audio 71.0% vs 64.9%, $p = 0.053$; fusion 84.1% vs 76.3%, $p = 0.037$). When men and women were compared for each emotion separately, 2 of the 12 comparisons (6 emotions $\times$ 2 models) reached $p < 0.05$ (fear and sadness from the voice), but none remained significant after correcting for the number of tests. With only 9 women and 10 men, these results should be seen as a weak tendency rather than a firm effect. Age was not significantly related to visual accuracy ($\rho = 0.26$, $p = 0.27$), audio accuracy ($\rho = -0.14$, $p = 0.56$) or disagreement ($\rho = -0.18$, $p = 0.46$). The test set had too few African American (3) and Asian (2) speakers to compare ethnic groups.

So the large differences between speakers seem to be mainly individual and are not clearly explained by gender or age. Each person has their own way of expressing emotions in the face and in the voice.

Table 9: Speaker-level averages by gender. p: Mann–Whitney U test over speakers.

	Female (9)	Male (10)	p
Visual accuracy (%)	69.2	66.9	0.71
Audio accuracy (%)	71.0	64.9	0.053
Fusion accuracy (%)	84.1	76.3	0.037
Disagreement (%)	49.1	53.2	0.44

3.6 Can disagreement information improve emotion recognition? (RQ5)

3.6.1 Step 1: Are the confidence values reliable?

To use the confidence of the models, their probabilities must be trustworthy. A model is *well calibrated* when, for example, its predictions made with 80% confidence are correct about 80% of the time. Figure 8a shows that both final models were already well calibrated. The audio model had an average confidence of 68.2% and an accuracy of 67.8% (ECE 0.02); the visual model had an average confidence of 71.2% and an accuracy of 68.0% (ECE 0.04). This was not the case for the first audio model, which was strongly overconfident (ECE 0.27, Section 3.1). Data augmentation and label smoothing therefore not only reduced overfitting but also made the audio model's confidence reliable.

We also tested temperature scaling, fitted on the validation set. For the audio model, it hardly changed anything (temperature $T = 1.12$). For the visual model ($T = 2.14$), it made the model underconfident on the test set (average confidence 54.4% for an accuracy of 68.0%; ECE 0.14). Calibration was therefore not necessary for the final models.

3.6.2 Step 2: Combining the two models

Figure 8b and Table 10 compare the different ways of combining the two models. All methods were fitted on the validation speakers and tested once on the test speakers.

Even simple rules worked well. Because both models gave reliable confidence values, simply trusting the more confident model reached 77.8%, 9.8 points better than the visual model. Multiplying the probabilities of the two models, which needs no training at all, reached 80.7%. Temperature scaling did not improve these rules; for the most-confident rule it even lowered accuracy to 75.8%, because the visual model became underconfident.

The best method used disagreement information. The best method was **stacking**, which learns from the validation set how to use both models' probabilities, their confidence and whether they agree. It reached **81.8%** accuracy and a macro-F1 of 0.815. This is 13.8 points better than the best single model (95% CI: +9.4 to +17.9 points; McNemar $p < 0.001$). Compared with the visual model, stacking corrected 299 wrong predictions and introduced 85 new errors.

Where does the gain come from? On clips where the two models agreed, there was little to improve: the single models were already right for 94.7% of these clips. The whole gain came from the clips where the models disagreed. On these clips, the visual model alone was right 42.6% of the time, while stacking was right 70.8% of the time (Table 10). In other words, the fusion learned which model to trust when they disagree.

Table 10: Fusion methods on the test set. Gain: accuracy gain over the visual model in percentage points, with 95% bootstrap confidence interval over speakers. p: McNemar test against the visual model. Disagr.: accuracy (%) on the 795 clips where the models disagreed.

Method	Macro-F1	Gain	p	Disagr.
Visual only	67.3	–	–	42.6
Audio only	67.7	-0.3	0.91	42.1
Most confident	77.5	$+9.8$ ($+6.2$, $+13.3$)	< 0.001	61.8
Most confident (cal.)	75.7	$+7.8$ ($+3.4$, $+12.2$)	< 0.001	57.9
Per-class trust rule	71.3	$+5.1$ ($+2.2$, $+8.1$)	< 0.001	52.6
Product	80.5	$+12.7$ ($+9.6$, $+15.7$)	< 0.001	67.4
Weighted average	79.8	$+12.0$ ($+8.3$, $+15.6$)	< 0.001	66.0
Stacking	**81.5**	$+13.8$ ($+9.4$, $+17.9$)	< 0.001	**70.8**
Upper bound	89.6	$+21.6$	–	84.8

Not every use of disagreement helped equally. A simple hand-made rule ("when the models disagree, trust the model that is more precise for the predicted emotion") gave only 73.1%, the weakest of all fusion methods. Learning the combination from data worked much better than fixed rules.

Robustness check. To check that these results do not depend on our particular validation speakers, we repeated the whole procedure with leave-one-speaker-out cross-validation on the test set. The results were very similar: 82.0% for stacking, 79.4% for the weighted average and 80.7% for the product. The upper bound in Table 10 (89.6%) is the accuracy that would be reached if the correct model were always chosen; it shows that there is still room for better fusion methods.

4. Discussion

This study looked at when and why an audio model and a visual model disagree when they recognise emotions. Below we discuss what our results mean, how they relate to earlier work, and what their limitations are.

4.1 Model disagreement is not the same as a conflict between voice and face

Our models disagreed on about half of the test clips. It would be easy to read this as "the voice and the face often show different emotions". The human ratings show that this is only partly true. Humans also saw different emotions in the voice and in the face for 42% of the clips, so real voice–face conflict is common, in line with research on hu-

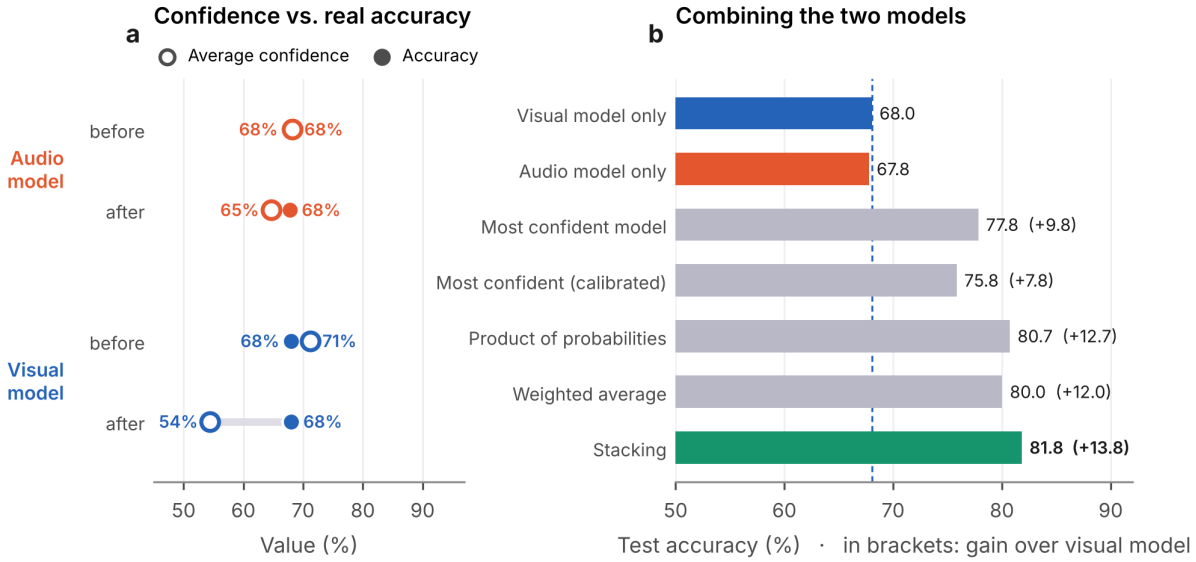


Figure 8: (a) Average confidence (open circles) and real accuracy (filled circles) of the two models on the test set, before and after temperature scaling. For a well-calibrated model, the two values are equal. (b) Test accuracy of the single models and of the fusion methods. Numbers in brackets show the gain over the visual model. The dashed line marks the visual model.

man perception (5, 6). However, only 47% of the model disagreements happened on such clips. The other 53% happened where humans saw the same emotion in both channels, so one of the models simply made a mistake. Model disagreement is therefore a mixture of two things: real differences between the channels and errors of the models. Studies that use the disagreement of separate models as a sign of emotional conflict should keep this in mind.

The type of disagreement also depended on the emotion. For disgust and happiness, most model disagreements were real conflicts. People who only listened often could not hear these emotions, while people who only watched could clearly see them. For neutral and sadness, most model disagreements were model errors.

4.2 The voice and the face carry different information

No single modality was best for all emotions. The face was clearly better for happiness and disgust, which are shown by strong facial movements such as a smile or a wrinkled nose. The voice was clearly better for sadness and neutral, and somewhat better for anger. This supports the idea that the two channels complement each other (3, 4), but our results show that this complementarity is not equal for all emotions. Combining the two models helped most for anger and fear, where neither modality was clearly better in F1. It helped very little where one modality already did most of the work.

An interesting difference appeared between the models and the human raters. For humans, the face was more informative than the voice for every emotion. For our models, the voice was better for sadness, neutral and anger. The audio model recognised two thirds of the sad clips from the voice, while humans who only listened recognised only one quarter. The model may use vocal cues, such as low energy or slow speech, that people do not rely on when they judge emotions. Because the test speakers were never seen during training, and every sentence was spoken with every emotion, these cues cannot come from the speaker's identity or from the words.

We also found that both channels become more useful when an emotion is expressed strongly, but the face benefits more (from 56% at low intensity to 79% at high intensity) than the voice (from 51% to 68%). For subtle emotions, the voice may therefore be as important as the face.

4.3 Each speaker is different

The results varied strongly between speakers. Some people were recognised much better from the face, others from the voice, and the two were unrelated. These differences were not clearly explained by gender or age; women only showed a weak tendency to be recognised better from the voice. This suggests that people have individual styles of expressing emotions. For a recognition system, this means that the best way to combine the voice and the face may differ from person to person. A system that adapts to the user could therefore work better than one fixed combination.

4.4 Disagreement is a useful signal

Agreement between the two models was a strong sign of a correct prediction: when they agreed, they were right 95% of the time. When they disagreed, a learned fusion method could decide which model to trust much better

than a fixed rule. This raised accuracy from 68.0% to 81.8%, which is higher than human raters who saw and heard the clips (76.5%). Two points were important. First, the confidence values of the models had to be reliable. Our first audio model was strongly overconfident (12); data augmentation and label smoothing solved this, so that even the simple rule of trusting the more confident model worked well. Second, learning the combination from data worked better than hand-made rules. Because the two models were trained separately, neither modality could dominate the training, a problem that has been reported for networks trained on several modalities together (7, 8).

4.5 Limitations

Our study has several limitations:

- **Small number of speakers.** The test set contained only 19 speakers, 9 women and 10 men. Small effects of gender or age could therefore not be reliably detected. The results for single speakers and emotions are based on only about 14 clips each and should be read with care.
- **Acted emotions.** CREMA-D contains emotions acted by actors, using 12 fixed sentences, in a controlled setting. Real, spontaneous emotions are often weaker and more mixed, so the results may differ in everyday situations.
- **One model design.** We used ResNet18 for both modalities to make the comparison fair. Other models may show different patterns. Both models still fitted the training data better than the validation data, although data augmentation reduced this problem for the audio model.
- **Simple pre-processing.** The audio spectrograms were normalised for each recording, which removes information about loudness. The face detector was simple and may have missed some faces.
- **Use of the validation set.** The validation set was used to choose the models (including the version of the audio model) and to fit the fusion. This may make the results slightly optimistic. However, all final results were measured on separate test speakers, and a leave-one-speaker-out check gave very similar results.
- **Comparison with humans.** The models were trained on the intended emotions, while human raters were not. Human ratings were also reduced to the most common answer for each clip. The comparison between models and humans is therefore not completely equal.

4.6 Future work

Several directions follow from this work:

- Repeating the analysis with **other models**, such as EfficientNet or transformer-based models for audio spectrograms and faces, to check whether the patterns hold.
- Using the **human ratings for training**, for example as soft labels, so that the models learn when the voice and the face really conflict.
- Developing **speaker-adaptive fusion**, which learns for each person how much to trust the voice and the face.
- Using disagreement to let a system **abstain** when it is unsure. When both models agreed, accuracy was 95%.
- Testing the approach on **spontaneous, real-world emotion data**.

5. Conclusion

We trained two emotion recognition models with the same design, one for the voice and one for the face, and studied when and why they disagree. Although the two models were almost equally accurate, they disagreed on about half of the CREMA-D test clips. Human ratings showed that about 47% of these disagreements reflect a real difference between what the voice and the face express, while the rest are errors of one of the models. The more useful modality depended on the emotion: the face was better for happiness and disgust, the voice for sadness, neutral and anger, while fear depended on the speaker. Each speaker also had an individual pattern that was not clearly explained by gender or age. Finally, disagreement turned out to be useful information: a learned fusion that uses the models' confidence and their disagreement improved accuracy from 68.0% to 81.8%, higher than human raters who both saw and heard the clips. Disagreement between voice and face should therefore not only be seen as a problem, but as a signal that can help multimodal emotion recognition.

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